MANTHAN
NEET 2027

ZOOLOGY | Lecture: 01

Body fluids and Circulation

Manish Sir

ZOOLOGY • MASTER TEACHER

TODAY'S GOAL

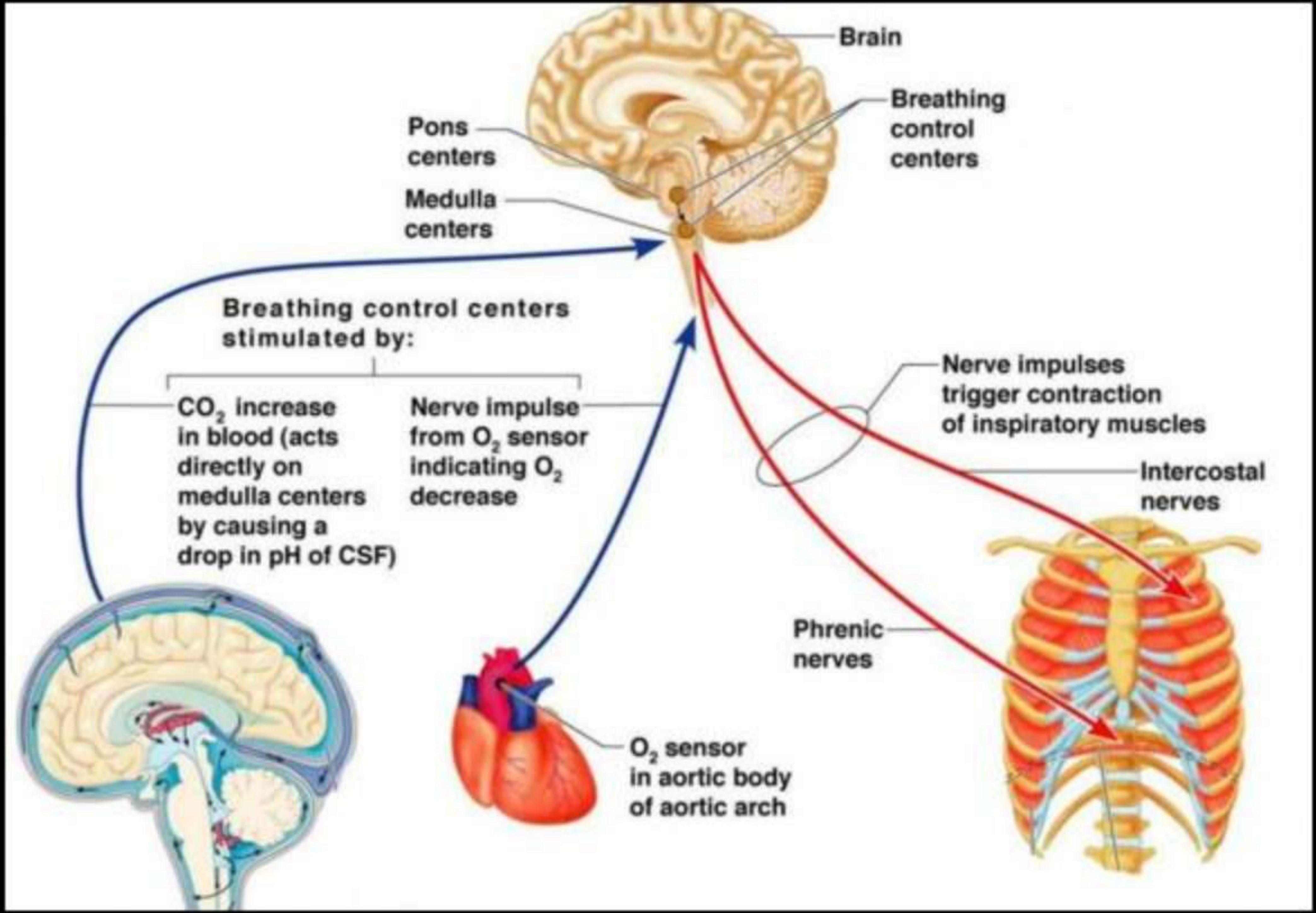
- Breathing Disorders and PYQs
- Blood ✓

Student Details Form

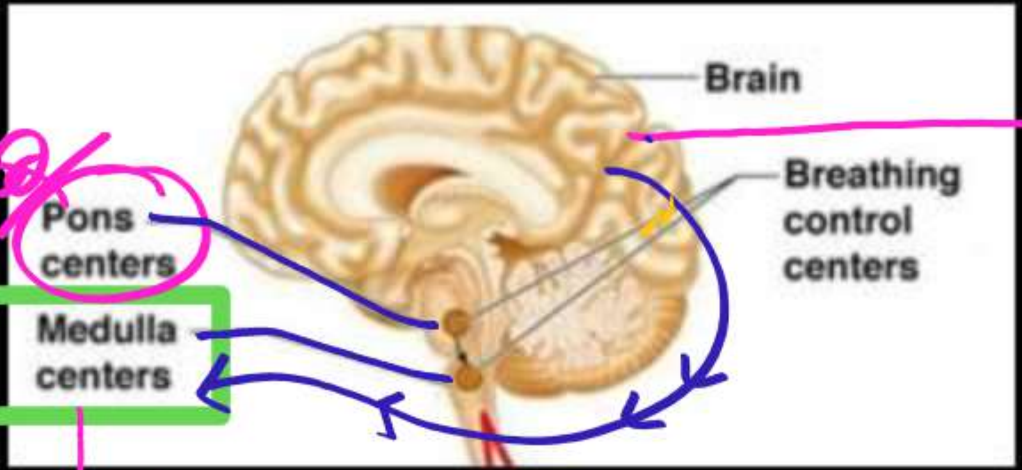
Telegram Channel

Dear students, Sacn the QR codes to fill mandatory Student Details Form and join the official Telagram channel.

Regulation of Breathing



Nervous Regulation of Breathing



Cerebrum → Voluntary Breathing

Medulla centers → Involuntary Breathing

M.O.

V.R.G = Ventral Respiratory Group

- Expiratory Area.
- Control Contraction of
  - I.I.C.M
  - Abdominal Mus.
- Control Forceful Breathing

D.R.G. - Dorsal - Respiratory - Group (Main)

- Inspiratory Area
- Control Contraction in
  - EICM
  - Diaphragm Mus.
- Control Normal Breathing
- Respiratory Rhythm Center

Pons Center = Pneumotaxic Center (Switch off Center)

Stop inspiration (Deep)

↓

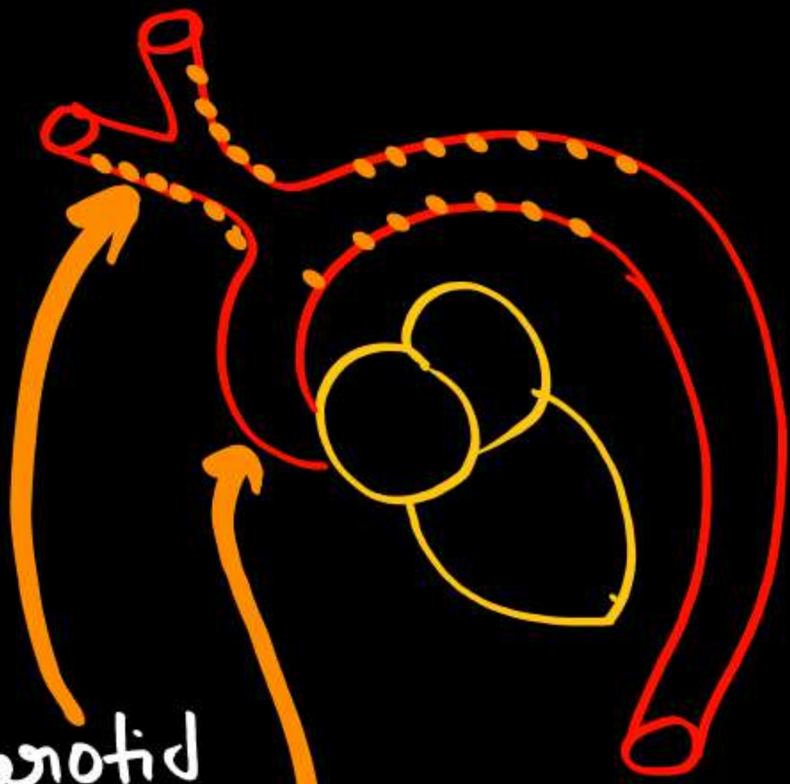
a/ ↓- Duration of Inspiration.

⇓

b/ [↑- Rate of Breathing / minute]

Chemical Regulation of Breathing

↳ Based on = Chemoreceptor, Sensitive for
- H^+
- CO_2



Center Chemoreceptor

[+nt in Brain Near M.O.]

[if H^+ & CO_2 ↑ in CSF]

Peripheral chemo.

[+nt on
- Carotid artery
- Aorta

[if ↑ H^+ & CO_2 in Oxy. Blood]

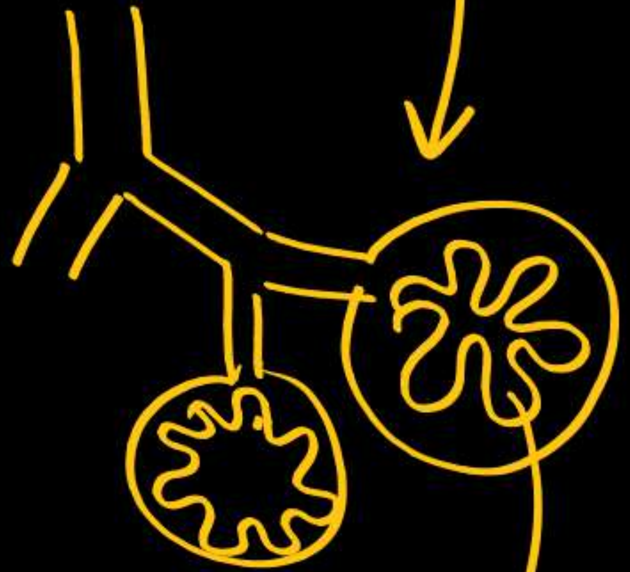
Cerebro Spinal Fluid

M.O. → ↑ - Rate of Breathing

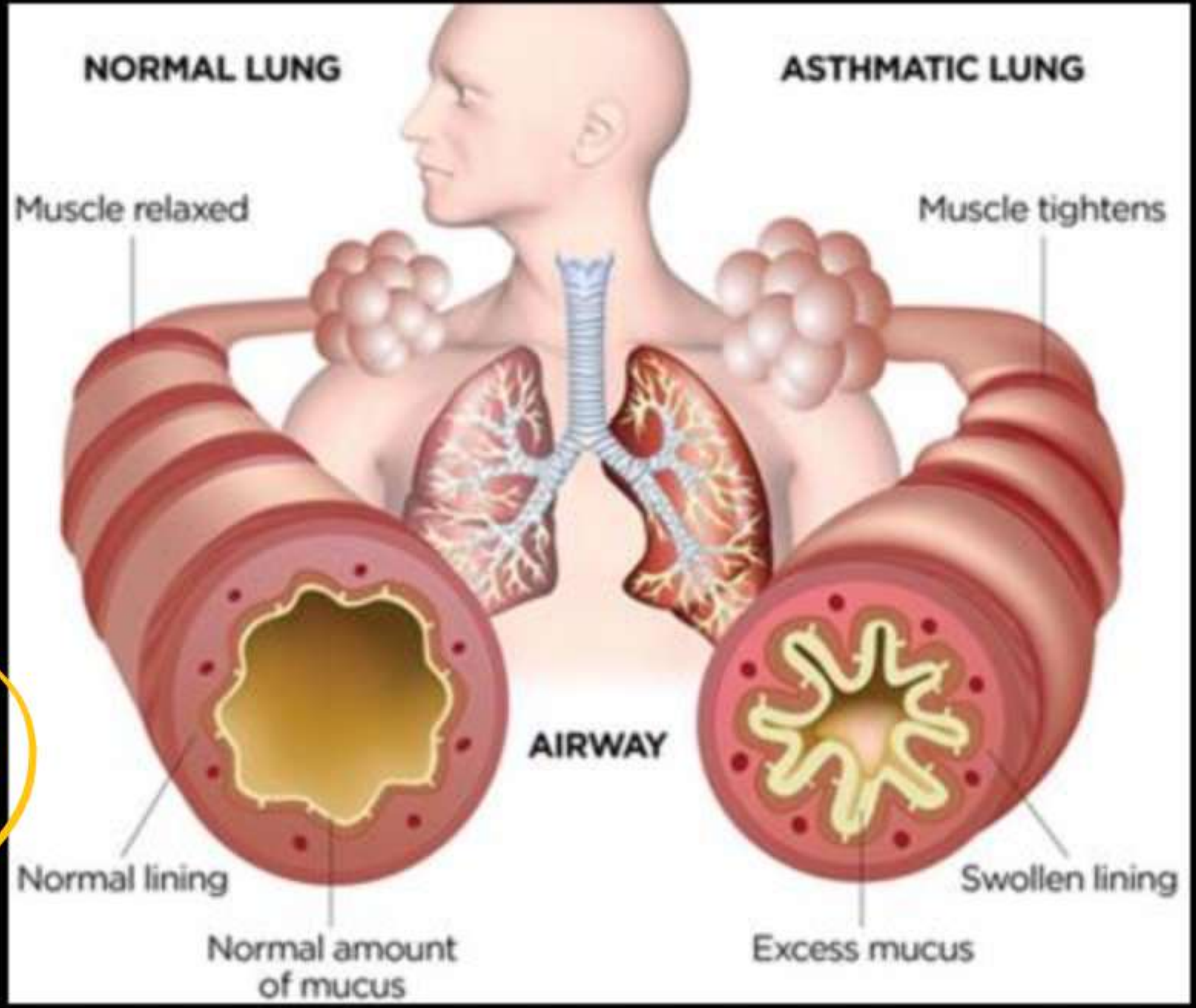
DISORDERS OF RESPIRATORY SYSTEM

(1) **Asthma** is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

Allergy



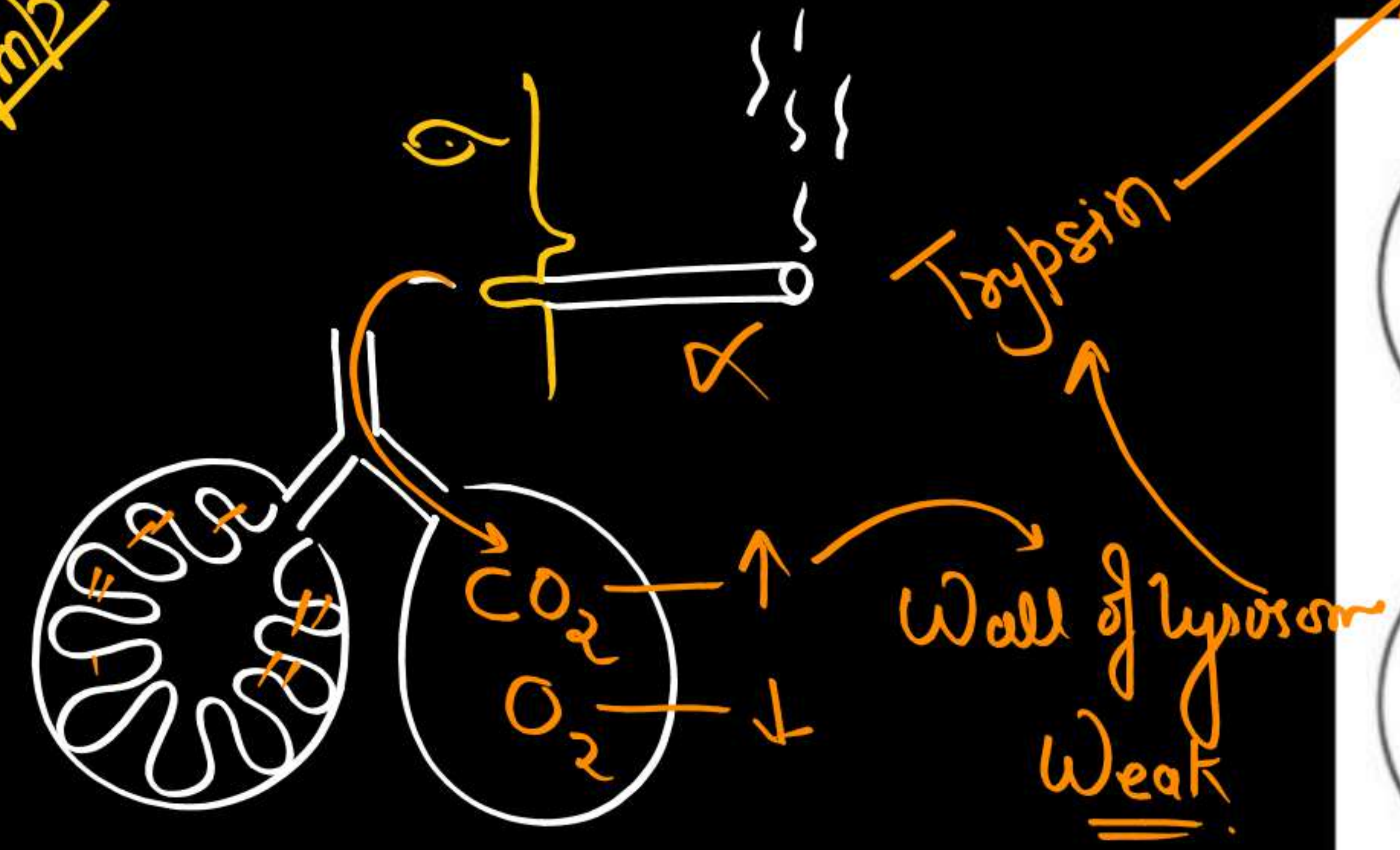
Bronchioconstriction

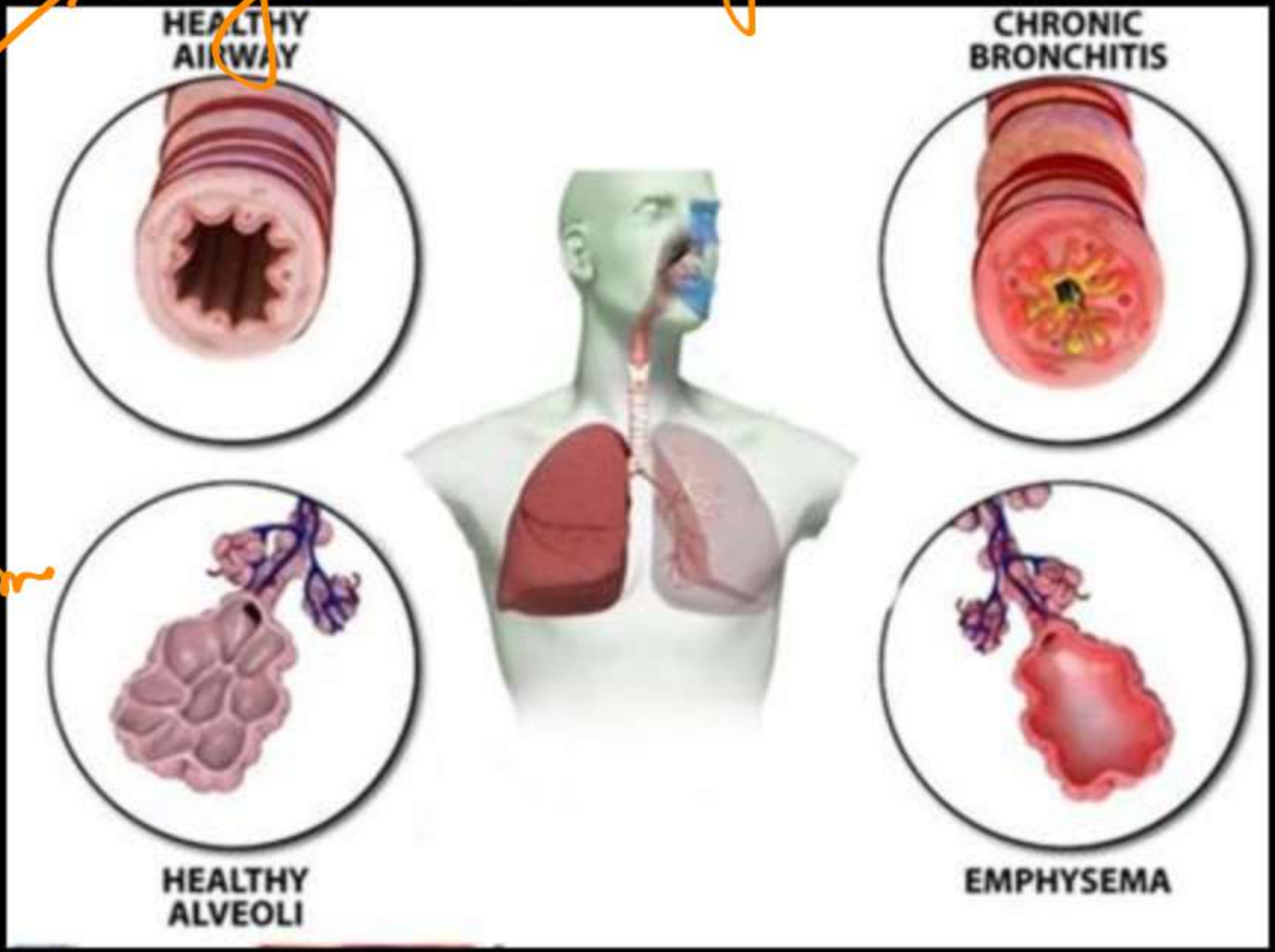


(2) **Emphysema** is a **chronic disorder** in which alveolar walls are damaged due to which **respiratory surface is decreased**. One of the major causes of this is **cigarette smoking.**

vvvImp

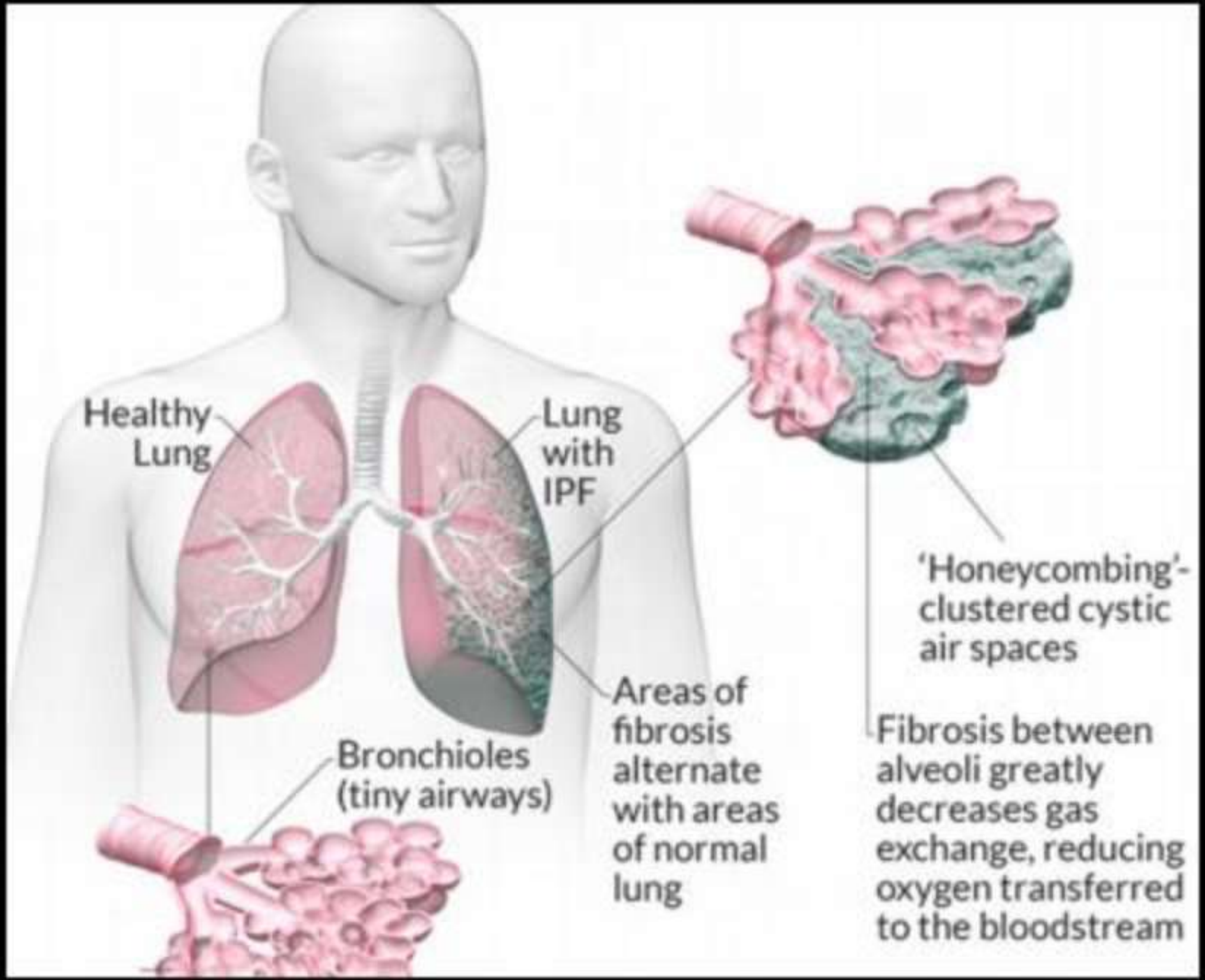
Digest Steptum of Alveoli





(3) Occupational Respiratory Disorders:

In certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation. Long exposure can give rise to **inflammation leading to fibrosis (proliferation of fibrous tissues) and thus causing serious lung damage.** Workers in such industries should wear protective masks.



Pneumoconiosis - Pneumoconiosis is a term that can refer to any chronic lung condition caused by the inhalation of dust or toxic fibers.

Coal worker Pneumoconiosis – Caused by inhalation of coal dust.

Silicosis - Caused by the inhalation of silica dust

Asbestosis - Caused by the inhalation of asbestos.

Chronic

Byssinosis - Caused by the inhalation of cotton fibers.

(4) Altitude Sickness/ Mountain Sickness-

High Altitude →
- Partial Pressure of O_2 ↓ (% of O_2 Same)
- $Hb-O_2$ Association = ↓
- $Hb-O_2$ - Dissociation = ↑

Hypoxia → Dizziness, Nausia, Vomiting, Breathlessness, ↑ - Rate of Breathing, Headach. ↑ - Heart Rate.

Stimulate → J.G. Cells of Kidney

~~Stomp~~

Release

✓ Erythropoietin Hormone → ↑ No. of R.B.C in Blood. [Erythropoiesis]

PYQ

1. Match **List - I** with **List – II**

2025

List - I	List - II
A. Emphysema	I. Rapid spasms in muscle due to low Ca^{++} in body fluid
B. Angina Pectoris	II. Damaged alveolar walls and decreased respiratory surface
C. Glomerulo-nephritis	III. Acute chest pain when not enough oxygen is reaching to heart muscle
D. Tetany	IV. Inflammation of glomeruli of kidney

Choose the **correct** answer from the options given below :

(a) A-II, B-IV, C-III, D-I

(b) A-II, B-III, C-IV, D-I

(c) A-III, B-I, C-IV, D-II

(d) A-III, B-I, C-II, D-IV

b

Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

(NEET 2024)

(a) High pO_2 and lesser H^+ concentration ✓

(b) Low pCO_2 and high H^+ concentration ✗

(c) Low pCO_2 and high temperature ✗

(d) High pO_2 and high pCO_2 ✗

a

Match List I with List II: (NEET 2024)

List I	List II
A. Expiratory capacity	I. Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
B. Functional residual capacity	II. Tidal volume + Expiratory reserve volume
C. Vital capacity	III. Tidal volume + Inspiratory reserve volume
D. Inspiratory capacity	IV. Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below:

(a) A-III, B-II, C-IV, D-I
(b) A-II, B-I, C-IV, D-III
(c) A-I, B-III, C-II, D-IV
(d) A-II, B-IV, C-I, D-III

d

Vital capacity of lung is______. (NEET 2023)

(a) IRV + ERV + TV + RV

(b) IRV + ERV + TV - RV

(c) IRV + ERV + TV

c

(d) IRV + ERV

Which of the following is not the function of the conducting part of the respiratory system?

(NEET 2022)

(a) It clears inhaled air from foreign particles

(b) Inhaled air is humidified

(c) Temperature of inhaled air is brought to body temperature

(d) Provides surface for diffusion of O_2 and CO_2

Under normal physical conditions in human beings every 100 ml oxygenated blood can deliver 5 ml of O_2 to the tissues. (NEET 2022)

(a) 2 ml

(b) 5 ml

(c) 4 ml

(d) 10 ml

Assertion (A): A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R): Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

(NEET 2021)

(a) If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If both assertion and reason are false.

a

Select the favorable conditions required for the formation of oxyhaemoglobin at the alveoli.

(NEET 2021)

(a) High pO_2, low pCO_2, less H^+, lower temperature a

(b) Low pO_2, high pCO_2, more H^+, higher temperature

(c) High pO_2, high pCO_2, less H^+, higher temperature

(d) Low pO_2, low pCO_2, more H^+, higher temperature

High pO_2, low pCO_2, lesser H^+ concentration and lower temperature are all favorable factors for the formation of oxyhaemoglobin at the alveoli.

The partial pressures (in mmHg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are: (NEET 2021)

(a) pO_2=104 and pCO_2=40

(b) pO_2=40 and pCO_2=45

(c) pO_2=95 and pCO_2=40

(d) pO_2=159 pO_2=159 and pCO_2=0.3

a

Select the correct events that occur during inspiration.

(NEET 2020)

(i) Contraction of diaphragm 1

(ii) Contraction of external intercostal muscles 2

(iii) Pulmonary volume decreases x

(iv) Intra pulmonary pressure increases x

(a) (i),(ii) and (iv) (b) Only (iv)

(c) (i) and (ii) (d) (iii) and (iv)

C

Identify the wrong statement with reference to transport of oxygen. (NEET 2020)

(a) Higher H^+ conc. in alveoli favors the formation of oxyhaemoglobin

(b) Low pCO_2 in alveoli favors the formation of oxyhaemoglobin

(c) Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2

(d) Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.

a

The Total Lung Capacity (TLC) is the total volume of air accommodated in the lungs at the end of a forced inspiration. This includes: (NEET 2020)

(a) RV; IC (Inspiratory Capacity); EC (Expiratory Capacity); and ERV

(b) RV; ERV; IC and EC

(c) RV; ERV; VC (Vital Capacity) and FRC (Functional Residual Capacity)

(d) RV (Residual Volume); ERV (Expiratory Reserve Volume); TV (Tidal Volume); and IRV (Inspiratory Reserve Volume)

d

Match the items given in Column I with those in Column II and select the correct option given below: **(NEET 2020)**

Column I		Column II	
a.	Pneumotaxic center	i.	Alveoli
b.	O_2 dissociation curve	ii.	Pons region of brain
c.	Carbonic anhydrase	iii.	Haemoglobin
d	Primary site of gas exchange	iv.	R.B.C.

(a) a-i, b-iv, c-ii, d-iii

(b) a-iii, b-i, c-iv, d-ii

(c) a-ii, b-iii, c-iv, d-i

(d) a-iv, b-iii, c-ii, d-i

c

Due to increasing air-borne allergens and pollutants, many people in urban areas are suffering from respiratory disorder causing wheezing due to:

(NEET 2019)

(a) inflammation of bronchi and bronchioles a

(b) proliferation of fibrous tissues and damage of the alveolar walls

(c) reduction in the secretion of surfactants by pneumocytes

(d) benign growth on mucous lining of nasal cavity.

T.V. + ERV

Tidal volume and expiratory reserve volume of an athlete is 500 mL and 1000 mL respectively. What will be his expiratory capacity of the residual volume is 1200 mL? **(NEET 2019)**

d

(a) 1700 mL (b) 2200 mL

(c) 2700 mL (d) 1500 mL

Select the correct statement. (NEET 2019)

(a) Expiration occurs due to external intercostal muscles

X

(b) Intrapulmonary pressure is lower than the atmospheric pressure during inspiration

✓ b

(c) Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.

X

(d) Expiration is initiated due to contraction of diaphragm.

X

The maximum volume of air a person can breathe in after a forced expiration is known as (NEET 2019)

(a) Expiratory Capacity

(b) Vital Capacity

(c) Inspiration Capacity

(d) Total Lung Capacity

b

Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively? (NEET 2018)

(a) Increased respiratory surface; Inflammation of bronchioles

(b) Increased number of bronchioles; Increased respiratory surface

(c) Inflammation of bronchioles; Decreased respiratory surface

c

(d) Decreased respiratory surface; Inflammation of bronchioles

Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of. (NEET 2017)

(a) Residual Volume

(b) Inspiratory Reserve Volume

(c) Tidal Volume

(d) Expiratory Reserve Volume

a

Name the chronic respiratory disorder caused mainly by cigarette smoking (NEET 2016)

(a) Emphysema (b) Asthma

(c) Respiratory acidosis (d) Respiratory alkalosis

Asthma may be attributed to (NEET 2016)

(a) Bacterial infection of the lungs

(b) Allergic reaction of the mast cells in the lungs

(c) Inflammation of the trachea

(d) Accumulation of fluid in the lungs

b

Reduction in pH of blood will (NEET 2016)

(a) Reduce the rate of heartbeat

(b) Reduce the blood supply to the brain

(c) Decrease the affinity of hemoglobin with oxygen

(d) Release bicarbonate ions by the liver

C

Body Fluids & Circulation

Circulatory system

1. Circulating fluid - Blood, Lymph
2. Blood vessels - Arteries, Capillaries, Veins and Portal System.
3. Pumping device – Heart

Blood

BLOOD

= Fluid Connective Tissue (origin - Mesodermal)

$$\uparrow CO_2 + H_2O \longrightarrow H_2CO_3 \longrightarrow \uparrow H^+ + HCO_3^-$$

- Volume of blood –
  - Male – = 5 L
  - Female – = 4.5 L
- ECF- = 15 L (Blood + Lymph + CSF + Synovial fluid etc.)
  - 15 L → 5 L (33% of ECF) = Blood
- pH of blood -
  - → 7.4 [oxy. blood] = $CO_2 \downarrow \longrightarrow H^+ \downarrow$
  - 7.2 [Deoxy. blood] = $CO_2 \uparrow \longrightarrow H^+ \uparrow$

Composition of Blood →

- Blood Plasma (55%)
  1) - Water → (90-92%)
  2) - Inorganic Sub. (1%)
  3) - Organic Sub. (8-9%)
- Formed Elements (Blood Cells) (45%)
  - Erythrocytes → Red → (R.B.C)
  - Leucocytes → White → (W.B.C)
  - Thrombocytes → Clote → (Platelets)

THANK YOU !!